ML Summer Bridge: Practice Problems

Machine Learning Class

Summer 2026

These problems are meant to reinforce the ideas from each week's notes. They will not be graded in the traditional sense (as in they won't count towards your GPA) but please turn them in by the first day of class. As we get closer to the start of the school year I will create a Google Classroom for our class where you may submit the assignment.

That said, these problems are for your own practice, and I'll happily talk through any of them with you once class starts. If you get stuck on something and want to check your thinking early, feel free to email me. Additionally, some of the questions may ask you to code things where you don't know the correct syntax (how to 'phrase things' / the correct command): feel free to use google or any other search engine to help you find the correct syntax / command, but please do not use AI. Using AI at this early stage would hinder your learning. Thank you.

Week 0 Problems

1. Open Google Colab and create a new notebook. In the first cell, import numpy and pandas using the abbreviations we discussed (np and pd), and print both of their version numbers. Write down what versions you get.

2. In your own words, explain why we write import numpy as np instead of just import numpy. What would change about the rest of our code if we left off the as np part?

3. Complete Units 3, 5, 7 and 9 on CodeHS if you haven't already. Then, without looking anything up, write a short Python program (on paper or in Colab) that:
   - stores your age in a variable,
   - uses an if-else statement to print "You can drive" if your age is 16 or older, and "You cannot drive yet" otherwise,
   - uses a loop to print the numbers 1 through 10,
   - and defines a function that takes in a number and returns that number squared.

4. If you have taken Java in 10th grade design engineering, write down two similarities and two differences you noticed between Python and Java while working through CodeHS.

5. Look back at the height table from the Week 0 notes. Just by looking at it (no code needed yet) - how many pieces of information do you think are recorded for each person? We'll formalize this idea next week, but see if you can guess the right vocabulary word for it.

Week 1 Problems

1. Using the original height table, write out the list representing the third data point, the way we did for the first data point in the notes.

2. Suppose our dataFrame data still has the original 5 rows and 4 columns. Without running any code, predict the output of each of the following, and then check yourself in Colab:
   - data.iloc[2, 3]
   - data.iloc[0, 1]
   - data.iloc[4, 3]
   - data.at[3, 'Gender']

3. Write a single line of code that would change the father's height of the person in row index 2 to 75.0.

4. Write a line of code using iloc that prints the entire fourth row of data. Then write a line using iloc that prints the entire third column.

5. After converting the Gender column so that M becomes 1 and F becomes 0, write down (by hand, no code) what the full table looks like.

6. In a sentence or two: why do you think it might cause problems if we assigned M to 1 and F to 2, instead of M to 1 and F to 0? (Remember - there's extra credit on the line for whoever can explain this well in our first class!)

7. What does data.shape return before and after you add a sixth row to the table with loc? Write both tuples.

Week 2 Problems

1. In your own words, explain the difference between a feature and a target. Give one example of each from our height table.

2. Suppose instead of predicting height, we wanted to predict a person's **gender** using father's height, mother's height, and height. Which column would now be the target? Which columns would be the features? Write out what the two new tables (features and target) would look like.

3. Write the two lines of code that would split our original data dataFrame into a features table X and a target column y, where Height is the target.

4. Why do you think we use a lowercase letter for y but an uppercase letter for X? (There's a real convention behind this that we'll explain properly in the fall - for now, just take your best guess.)

5. Write the code to produce a plot of Height as a function of Mother's Height. What trend, if any, do you notice?

6. Looking at both plots (Height vs. Father's Height and Height vs. Mother's Height) side by side, which relationship looks stronger to you? There's no strictly right answer here - just explain your reasoning.

Week 3 Problems

1. Make a scatter plot of Height vs. Father's Height. In a sentence, describe the shape of the cloud of points. Does it look roughly linear, or is the pattern harder to describe with a single straight line?

2. Suppose we meet someone whose mother is 66 inches tall, and we want to guess their height using only mother's height. Using your scatter plot from Week 3, draw a line by eye and use it to make a prediction. Write down your predicted height.

3. Have a friend or family member draw their own "by eye" line through the same scatter plot, and use it to make the same prediction from Problem 2. Did you get the same answer? Why do you think that is (or isn't) a problem?

4. Here's a thought experiment: instead of drawing a line at all, what if we just predicted *the same height for everyone*, say the average height in our table? Do you think this would generally do a better or worse job than using a line? Explain your reasoning in a few sentences.

5. Without doing any calculations, write down in your own words what you think it would mean for one line to be "better" than another line at fitting a set of data points. Keep this answer - we'll come back to it in very early on - and repeatedly throughout the year.